

DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 4 (AT1)

Chart Construction Test

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Question 1 (Healthcare Analytics)

Program

```
# Create the dataset
```

```
patient <- c("P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8")
```

```
age <- c(24, 32, 41, 48, 56, 65, 38, 29)
```

```
bmi <- c(19.8, 24.5, 28.1, 30.3, 33.2, 36.5, 26.2, 22.1)
```

```
blood_pressure <- c(110, 118, 132, 145, 152, 168, 124, 116)
```

```
severity <- c("Low", "Medium", "High", "High",  
             "Critical", "Critical", "Medium", "Low")
```

```
# Assign colors according to disease severity
colors <- c("green", "yellow", "orange", "orange",
           "red", "red", "yellow", "green")

# Create point sizes based on BMI values
sizes <- bmi / 2

# Create scatter plot
plot(age,
     blood_pressure,
     col = colors,
     pch = 19,
     cex = sizes,
     xlab = "Age",
     ylab = "Blood Pressure (mmHg)",
     main = "Healthcare Analytics Visualization")

# Add patient labels
text(age,
     blood_pressure,
     labels = patient,
     pos = 3,
     cex = 0.8)

# Add legend
legend("topleft",
     legend = c("Low", "Medium", "High", "Critical"),
```

```
col = c("green", "yellow", "orange", "red"),
pch = 19)
```

Output



Parameter	Representation
Age	X-axis
Blood Pressure	Y-axis
Disease Severity	Color
BMI	Point size

- The **x-axis** represents age because age strongly influences disease progression.
- The **y-axis** represents blood pressure because it is an important health indicator.
- **Colors** help doctors quickly identify the severity level.
- **Point size** represents BMI, allowing easy identification of obesity-related risks.

Question 2 (Climate Science)

Program

```
# Install the package if it is not available
install.packages("plotrix")
```

```

# Load the package
library(plotrix)

# Dataset
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

solar_radiation <- c(180, 200, 240, 280, 320, 300,
                    260, 250, 240, 220, 190, 170)

temperature <- c(20, 23, 29, 35, 39, 36,
                32, 31, 30, 28, 24, 21)

rainfall <- c(25, 18, 12, 8, 5, 65,
             150, 170, 120, 85, 45, 30)

# Divide the plotting area into two sections
par(mfrow = c(1, 2))

# ----- Cartesian Line Chart -----

plot(temperature,
     type = "o",
     xaxt = "n",
     xlab = "Month",
     ylab = "Temperature (°C)",
     main = "Temperature Variation")

```

```
axis(1,  
      at = 1:12,  
      labels = month)  
  
grid()  
  
# ----- Polar Chart -----  
  
radial.plot(rainfall,  
            labels = month,  
            main = "Annual Rainfall")  
  
# Return to the default layout  
par(mfrow = c(1, 1))
```

Output

- Exact values are harder to compare.
- Interpretation may be difficult for large datasets.

Interdisciplinary Integration

- **Environmental Science:** Studies weather patterns and climate changes.
- **Mathematics:** Uses Cartesian and polar coordinate systems.
- **Data Visualization:** Helps represent complex climate data effectively.

Question 3 (Business Analytics & Marketing)

Program

```
# Dataset
```

```
product <- c("A", "B", "C", "D", "E")
```

```
sales <- c(210, 185, 260, 145, 120)
```

```
profit <- c(18, 14, 25, 12, 10)
```

```
market_share <- c(24, 20, 28, 15, 13)
```

```
# Create a color scale according to profit percentage
```

```
colors <- c("lightblue", "skyblue", "dodgerblue",  
           "blue", "darkblue")
```

```
# Highlight the product with the highest market share
```

```
highest <- which.max(market_share)
```

```
colors[highest] <- "red"
```

```
# Create the bar plot
```

```

barplot(sales,
        names.arg = product,
        col = colors,
        main = "Sales, Profit, and Market Share Analysis",
        xlab = "Products",
        ylab = "Sales (₹ Lakhs)")

```

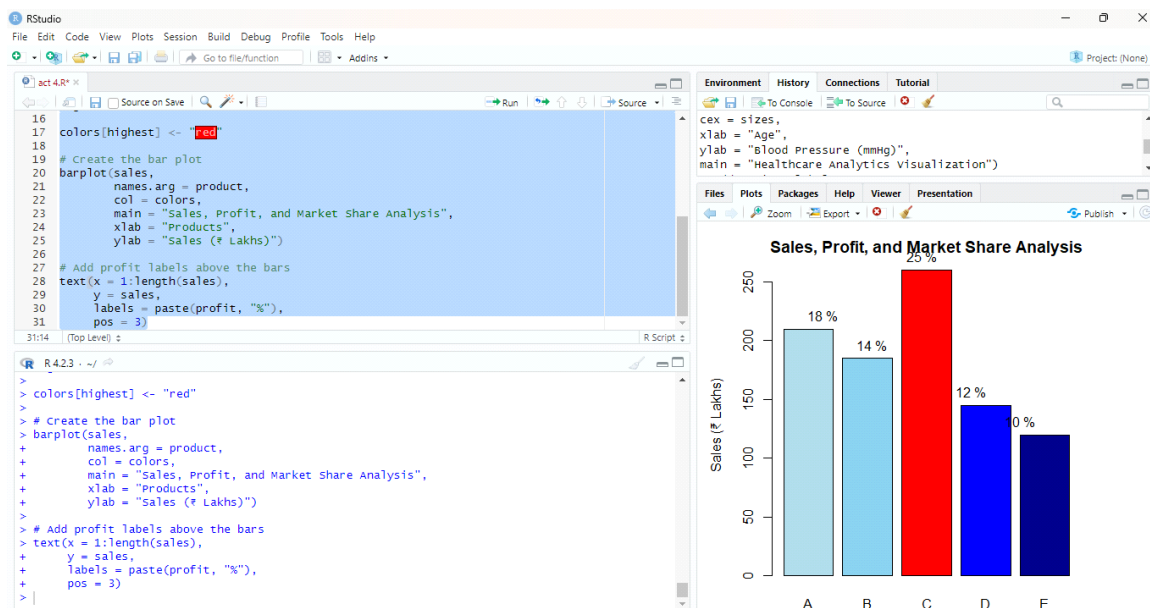
Add profit labels above the bars

```

text(x = 1:length(sales),
     y = sales,
     labels = paste(profit, "%"),
     pos = 3)

```

Output



Element	Representation
Bar height	Sales value
Color scale	Profit percentage
Red color	Highest market share

Labels	Profit percentage
--------	-------------------

Interpretation

- Taller bars indicate higher sales values.
- Darker colors represent higher profit percentages.
- The red bar highlights the product with the largest market share.
- Profit values are displayed above each bar.

Question 4 (Transportation Engineering)

Program

```
# Install the package if it is not available
```

```
install.packages("plotrix")
```

```
# Load the library
```

```
library(plotrix)
```

```
# Dataset
```

```
direction <- c("North", "North-East", "East", "South-East",  
              "South", "South-West", "West", "North-West")
```

```
vehicles <- c(980, 760, 1240, 890, 1420, 1130, 860, 720)
```

```
# Assign colors according to traffic density
```

```
colors <- c("lightgreen", "green", "yellow", "orange",  
           "red", "darkorange", "skyblue", "blue")
```

```
# Create the polar chart
```

```
radial.plot(  
  vehicles,  
  labels = direction,
```

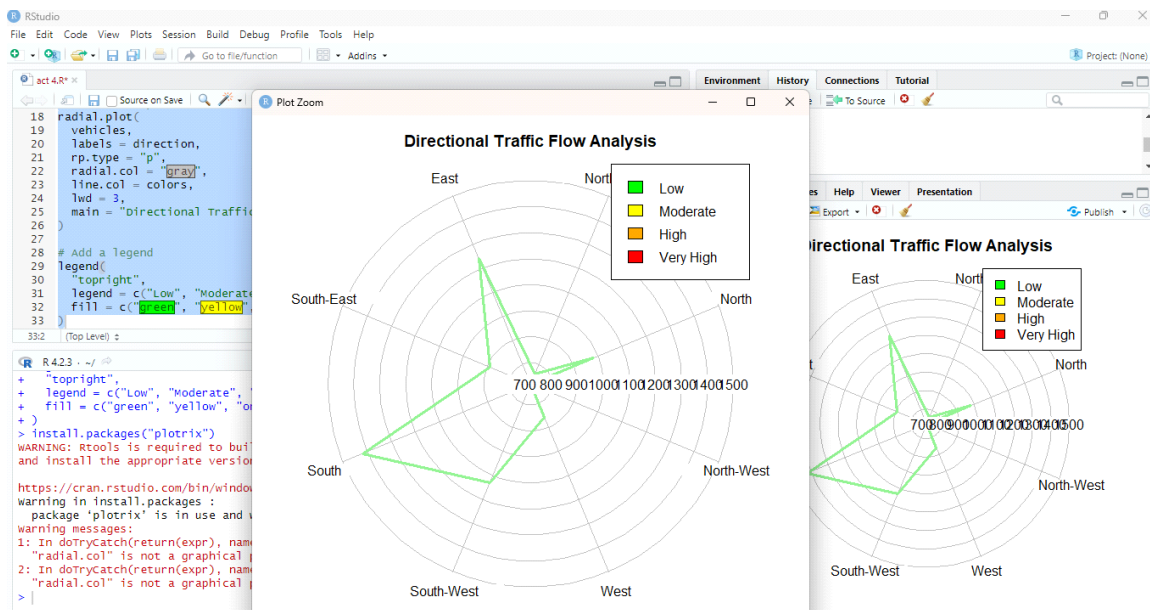
```

rp.type = "p",
radial.col = "gray",
line.col = colors,
lwd = 3,
main = "Directional Traffic Flow Analysis"
)

# Add a legend
legend(
  "topright",
  legend = c("Low", "Moderate", "High", "Very High"),
  fill = c("green", "yellow", "orange", "red")
)

```

Output



Polar Chart	Cartesian Bar Chart
Represents circular movement naturally.	Represents data in a straight line.
Displays directional traffic patterns clearly.	Makes directional analysis more difficult.
Shows the relationship between neighbouring	Treats directions as separate categories.

directions.	
Useful for transportation and GIS applications.	More suitable for simple comparisons.

Interpretation

- **South (1420 vehicles/hour)** shows the highest traffic density.
- **East (1240 vehicles/hour)** and **South-West (1130 vehicles/hour)** also experience heavy traffic.
- **North-West (720 vehicles/hour)** has the lowest traffic volume.

Question 5 (Environmental Sustainability Dashboard)

Program

```
# Create the dataset
```

```
city <- c("A", "B", "C", "D", "E", "F")
```

```
aqi <- c(48, 82, 135, 165, 72, 95)
```

```
green_cover <- c(45, 32, 21, 18, 38, 30)
```

```
co2 <- c(1.8, 3.5, 5.6, 7.4, 2.9, 4.2)
```

```
population <- c(9, 15, 22, 28, 12, 18)
```

```
# AQI colour categories
```

```
colors <- c("green", "yellow", "orange",  
           "red", "yellow", "orange")
```

```
# Point size based on population
```

```
size <- population / 3
```

```
# Create two plots in a single dashboard
```

```
par(mfrow = c(1, 2))
```

```
# ----- Plot 1 -----
```

```
plot(green_cover,  
     co2,  
     pch = 19,  
     cex = size,  
     col = colors,  
     xlab = "Green Cover (%)",  
     ylab = "CO2 Emission",  
     main = "Green Cover vs CO2 Emission")
```

```
text(green_cover,  
     co2,  
     labels = city,  
     pos = 3)
```

```
legend("topright",  
       legend = c("Good", "Moderate",  
                  "Poor", "Very Poor"),  
       fill = c("green", "yellow",  
                "orange", "red"))
```

```
# ----- Plot 2 -----
```

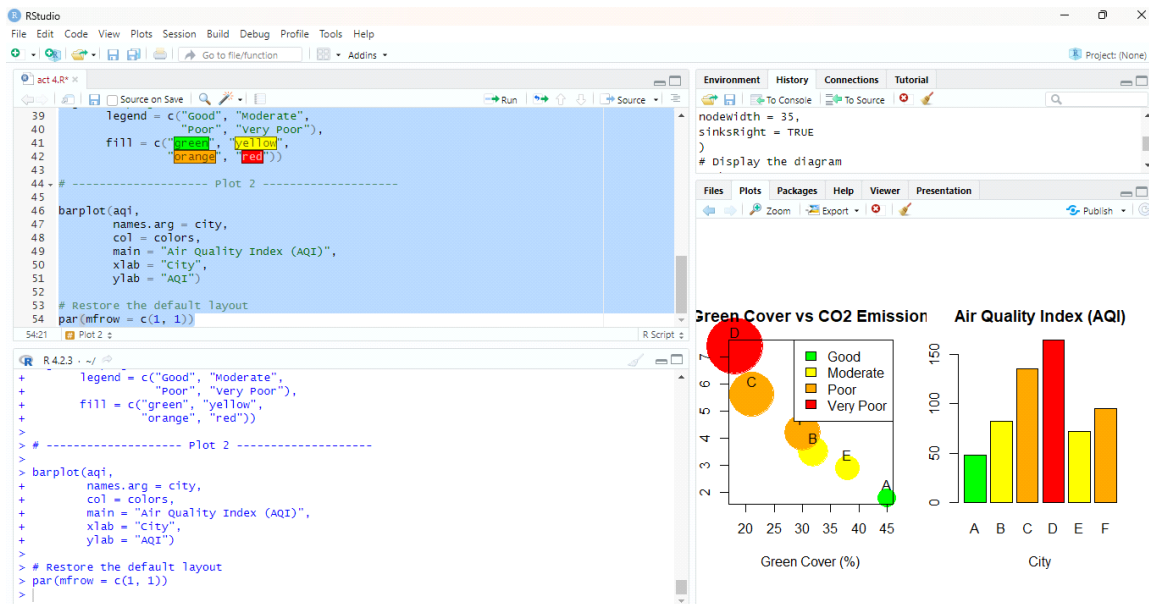
```
barplot(aqi,  
        names.arg = city,
```

```
col = colors,
main = "Air Quality Index (AQI)",
xlab = "City",
ylab = "AQI")
```

Restore the default layout

```
par(mfrow = c(1, 1))
```

Output



Justification of Design Choices

- **Scatter plot:** Shows the relationship between green cover and CO₂ emission.
- **Bar chart:** Allows easy comparison of AQI values among cities.
- **Colour scale:** Helps identify pollution levels quickly.
- **Point size:** Represents population density effectively.
- **Labels:** Improve readability and interpretation.

Interpretation

- City **D** has the highest AQI level and CO₂ emission.

- City **A** has the largest percentage of green cover and the best air quality.
- Cities with less green cover generally exhibit higher pollution levels.

This dashboard integrates **environmental science**, **sustainability analysis**, and **data visualization** to support environmental decision-making.